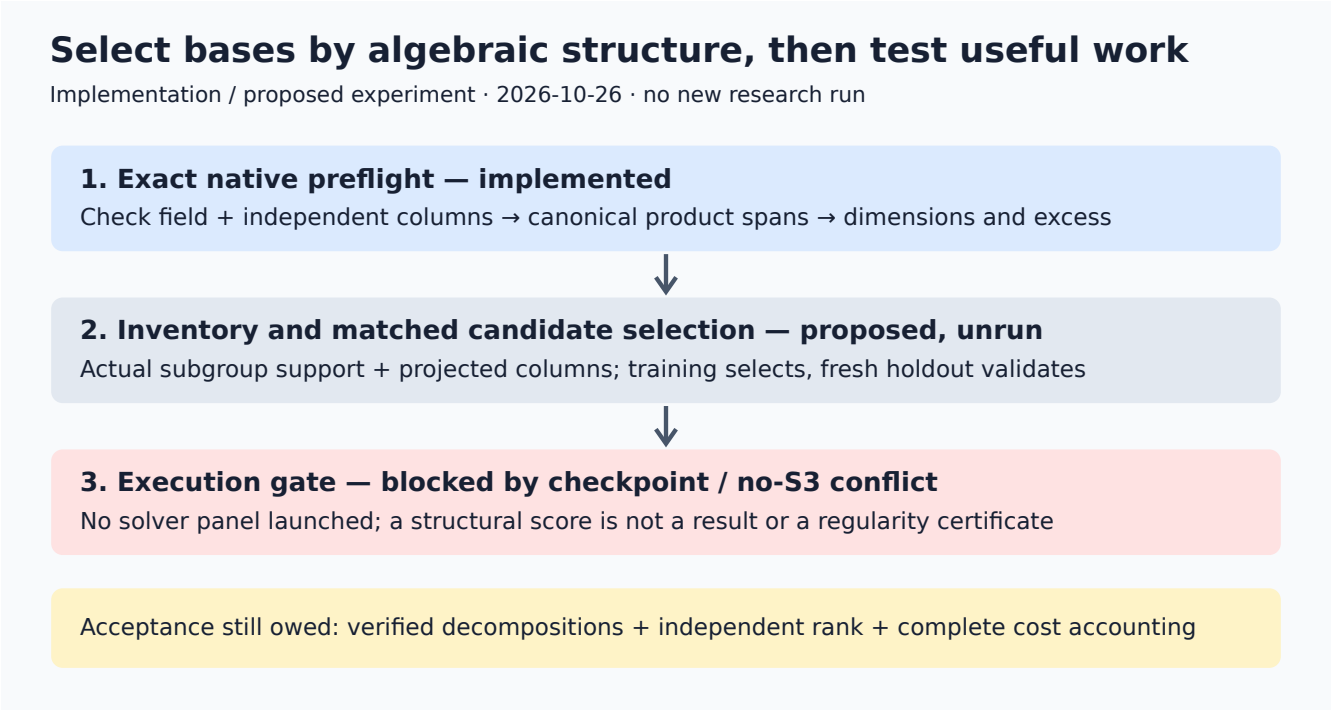
1. Factor-base product-profile screening

2026-10-26 · implementation and literature-led next experiment

Status: no new research panel executed; factor-base improvement unestablished.

The user requested continued work toward a useful factor base. The current lead atlas points to linearization excess as an unexamined transfer question. This change implements exact native structural profiling inside the existing `factor-base-sweep` CLI. It does not substitute a score for successful solving.



1.1. What the command computes

Input: an explicit binary field modulus and independent polynomial-basis columns of an abscissa subspace V. Output: canonical bases for product spans V, V^(2), … through the requested power, their dimensions, a basis-invariant subspace identity and the two-summand expression `e = n - dim(V) - dim(V^(2))`.

The field modulus is checked for irreducibility. Invalid, truncated and dependent inputs are rejected. Output is write-once. Wide field structure through degree 571 is supported; wide point decomposition is not added. The sum of product dimensions is not proof of independent symmetric variables or valid summand splitting. Product multiplication counts omit other work. Predicted degree, certified regularity and total-cost winner remain null.

1.2. Why this lead, and what is already known

The sibling PDP study reports a low-error empirical two-summand degree fit over 656 bases. That is historical evidence from its own solver, not a native measurement here or a rule for m=3/4. We read its definitions, fitting scope, projection policy and failure limits; none of its Python experiment code ran.

Bachoc–Serra–Zémor, Theorem 3, characterizes minimal product growth in finite prime-degree extensions under dimensions at least two and product dimension at most n-2. Geometric progressions are known structure in that range, not a new discovery. The theorem proves no solver-complexity or regularity bound. The full source/version record is in `LITERATURE.md`.

1.3. The pending experiment

Pair high-excess and median-excess candidates with random references. Use exact subgroup-qualified inventories and separately record x-counts, actual points and distinct signed projected columns. Equal subspace dimension alone does not match factor-base size. Training selects; fresh holdout tests the selection; conditioned controls never count as random yield.

Primary exploratory model: `icv1-f2m31-tm90707-c95f16f5`, r=1439393,h=1492. Secondary model: `icv1-f2m19-t797-b6cf2467`, r=130873,h=4. Registry field and generator records are named in `PROTOCOL_DRAFT.md`. Degree 31′s Frobenius module differs from 83/131; it cannot replace the required degree 83 gate.

The first native comparison is m=2. An m=3/4 follow-up needs a native adapter, valid exact descended systems and its own controls before a runnable protocol exists. No old XP card is promoted or dispatched.

Acceptance needs complete charged successful decomposition work and useful independent relations, including construction, scoring, failures, verification and rank. The same native-operation pricing band applies to every arm. Unemitted counters prevent a total-cost ranking. Fastest claims additionally owe matched external solvers. There is no end-to-end, rho or speed claim.

1.4. Concrete execution blocker

Current AGENTS requires an S3 resume checkpoint before an experiment starts and tells Codex not to launch if `scripts/checkpoint.sh gate` fails. The existing user restriction forbids S3 operations. The local gate returned 2 because no resume file/checkpoint exists, before any upload was attempted. No experiment ran. This is an authorization/policy conflict, not evidence of an unavailable mathematical route or a failed cloud credential.

The implementation and correctness validation can be reviewed now. New experiment execution requires checkpoint authorization or an explicit owner-approved alternative. The broader goal remains open.

1.5. Sources and evidence

- Native command: `src/bin/factor_base_sweep/profile.rs`.
- Current atlas: `LEAD-ICE-linearization-excess-fb-prior`.
- Sibling source: `cryptanalysis:experiments/pdp-degree-heuristics/README.md`.
- Primary theorem: arXiv 1501.00602v2, pp. 1–3; DOI 10.1017/S0305004117000044.
- Conservation scope: `crypto-autoresearcher:knowledge/findings/KN-FIND-007.md`.
- Prior native null: `research/dreg_closure_specialise_20261017/RESULTS.md`.
- Correctness/command outcomes: `VALIDATION.md` and its linked receipts.

No new measured solver result changes the canonical scoreboard, leaderboard, browser, benchmark copies or old sealed research. No new Lean proof or identity certificate is claimed for the cited existing mathematics.